CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (CE)

March 2018

CE741 Information & Network Security

Date: 27.03.2018, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Figures to the right indicate marks.

SECTION-I

Q-1	**Answer the following questions.**	
[A]	What is symmetric-key and asymmetric-key cryptography?	**4**
[B]	Explain Rail Fence Cipher with example.	**4**
[C]	Why non-linearity is required in cryptography system?	**4**
Q-2	**Answer the following questions.**	
[A]	What is block cipher in cryptography. Explain in details.	**4**
[B]	Explain linear cryptanalysis in brief.	**4**
[C]	Draw the process flow diagram of DES encryption algorithm.	**4**
	OR	
Q-2	**Answer the following questions.**	
[A]	What is steganography in security? Explain in brief.	**4**
[B]	Explain differential cryptanalysis in brief.	**4**
[C]	Draw the process flow diagram of AES encryption algorithm.	**4**
Q-3	**Answer the following questions.**	
[A]	Explain MAC in details.	**5**
[B]	Explain Hill cipher in details.	**6**

SECTION-II

Q-4 Answer the following questions.

[A] Explain *confusion* with respect to cryptography. **4**

[B] Explain any two RSA attacks. **4**

[C] Explain *affine cipher* with example **4**

Q-5 Answer the following questions.

[A] Explain *defusion* with respect to cryptography **4**

[B] Explain *Preimage* attack in detail. **5**

[C] What is one-way function in cryptography? Why it is used? **6**

OR

Q-5

[A] Explain *MixColumns* transformation of AES. **4**

[B] "The main idea behind symmetric-key cryptography is the concept of the trapdoor one-way function." – Justify the statement. **5**

[C] For p = 11 and q = 17 and choose e=7. Apply RSA algorithm where Cipher message=11 and thus find the plain text. **6**

Q-6 Answer the following questions.

[A] Explain the working of MD5 Hashing algorithm **4**

[B] Explain the concept of HMAC in detail **4**

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